

Jingxuan (Jasper) Wu

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Education

The University of North Carolina at Chapel Hill

M.S. in Statistics, Analytics, and Data Science

Aug. 2025 – Dec. 2026(expected)

Chapel Hill, NC, USA

The Chinese University of Hong Kong, Shenzhen

B.S. (Hons) in Data Science and Big Data Technology

Sep. 2021 – May 2025

Shenzhen, China

Selected Publications (* indicates equal contribution)

Conference & Workshop Papers

- [C1] **Jingxuan Wu**, Zhenglin Wan, Xingrui Yu, Yuzhe Yang, Bo An, Ivor Tsang, and Yang You. **“Letting Trajectories Spread: Quality-Preserving Control for Diverse Flow Matching”**. 2026.
Proceedings of the 43rd International Conference on Machine Learning (ICML), 2026.
- [C2] Zhenglin Wan, **Jingxuan Wu**, Xingrui Yu, Chubin Zhang, Mingcong Lei, Bo An, Ivor W. Tsang, and Yang You. **“Adversarial Dual On-Policy Distillation from Expressive Teacher”**. 2026.
Proceedings of the 43rd International Conference on Machine Learning (ICML), 2026.
- [C3] **Jingxuan Wu**, Zhenglin Wan, Yuzhe Yang, Yiqiao Huang, Chubin Zhang, Xingrui Yu, Ivor Tsang, and Yang You. **“Denoising Time Matters: Diverse Generation in Diffusion Language Models”**. 2026.
40th Conference on Neural Information Processing Systems (NeurIPS), 2026.

Preprints & Technical Reports

- [P1] **Jingxuan Wu**, Yuzhe Yang, Yiqiao Huang, Chengzhi Liu, Qingni Wang, Chengxuan Qian, Shutong Wu, Jiawei Zhang, and Xin Eric Wang. **“MemFold: Learning Compact Soft Memory for Long-Context Personalization via On-Policy Optimization”**. 2026. Under review at ICLR2026.
- [P2] Hanzhao Wang*, **Jingxuan Wu***, Yumeng Li, Yu Pan, and Guanting Chen. **“LLM-Powered Predictive Decision-Making for Sustainable Data Center Operations”**. arXiv preprint arXiv:2608.18503. 2026.

Research Experience

Department of Computer Sciences, University of Wisconsin–Madison

Advisor: [Prof. Jiawei Zhang](#)

May 2026 – Present

Madison, WI, USA

- **Generalizable Memory Management for Cross-Dataset Reasoning**: Identified substantial cross-dataset degradation in existing task-specific memory methods and developed a unified memory-management framework trained across heterogeneous benchmarks. The framework aims to learn transferable memory policies that can be directly deployed on unseen OOD datasets without additional training.
- **MemFold: Compact Soft Memory for Long-Context Personalization**
 - Proposed MemFlow, which compresses query-conditioned textual memory into a fixed budget of K soft tokens and optimizes the reader on its own rollouts using group-relative task rewards and confidence-gated on-policy distillation from a frozen textual-memory teacher. The teacher is never sampled and is removed entirely at inference.
 - Achieved the highest accuracy across PersonaMem-32K/128K, PrefEval, and LongMemEval with three Qwen backbones, improving weighted-average accuracy by **7.7–10.6 percentage points** over the strongest competing baselines while using approximately **10% fewer end-to-end tokens** than full-context inference and transferring to two unseen benchmarks without target-domain training.

A*STAR Centre for Frontier AI Research

Advisors: [Prof. Xingrui Yu](#), [Prof. Ivor Tsang](#), [Prof. Bo An](#)

May 2025 – Present

Remote & Singapore

- **TAPS: Training-Free Diversity Control for Diffusion Language Models**

- Proposed a training-free sampling method that applies time-annealed, manifold-constrained perturbations to context embeddings during early denoising.
- Showed that TAPS improves meaningful reasoning-path diversity rather than merely surface-level variation. On correctness-controlled subsets of GSM8K and HumanEval+, TAPS outperformed all evaluated sampling baselines across every structural metric, increasing distinct reasoning paths by **20.9%** and normalized-AST diversity by **26.8%** over the strongest baselines while reducing identical correct solutions by **43.1%** and **20.2%**, respectively.
- **OSCAR: Quality-Preserving Diversity Control for Text-to-Image Flow Models**
 - Proposed a set-level feature-volume objective with redundancy-aware reweighting to spread predicted endpoints in semantic space without retraining the base model.
 - Derived orthogonal stochastic controls that increase diversity surrogate without opposing the base generative flow, improving semantic diversity by **14.8%** and pixel-level diversity by **14.1%** while reducing KID by **11.8%** on compositional T2I-CompBench prompts and preserving prompt alignment.
- **FA-OPD: Dual On-Policy Distillation for Efficient Embodied Control**
 - Proposed a teacher–student framework that co-trains an expressive Flow-Matching teacher to supervise a lightweight MLP policy on its own rollouts through complementary reward and action distillation, balancing long-horizon exploration with dense local correction.
 - Validated FA-OPD across six navigation, manipulation, and locomotion benchmarks, achieving the best final performance in five of six environments while converging faster and with lower variance than strong behavioral-cloning and adversarial-imitation baselines. Demonstrated stronger generalization to out-of-distribution states and robustness under noisy, limited, or sub-optimal demonstrations, while retaining single-pass MLP deployment.

Department of Statistics & Operations Research, UNC-Chapel Hill

Advisors: Prof. Guanting Chen, Prof. Xiaocheng Li, Prof. Wenjia Ba

Jan. 2024 – Present

Remote & Chapel Hill, NC

- **SafeScreen: Bias-Safe AI Screening for Adaptive Visual Conjoint**
 - Proposed SafeScreen, which confines AI feedback to acquisition while fitting the preference model exclusively on trusted human comparisons.
 - Built a continuous product-design space in text-to-image embedding space and developed a Thompson-sampling procedure that screens B candidates into m human queries while reserving one unscreened selection. Established bias-safe inference under arbitrary AI screening and up to a B/m improvement in human-label efficiency under calibrated AI–human alignment.
- **LLM-DC: Predictive Scheduling for Sustainable Data Centers**
 - Developed an LLM-powered scheduling system that predicts GPU runtime and energy consumption from source code and uses a value-based online allocator to optimize GPU assignment, reducing energy consumption by **31.6%** and waiting time by **30.3%** in backtesting on real data-center operational traces.

Projects

CaveAgent | Python, LLM, AST, LiteLLM, Asyncio, Security Validation

Oct. 2025 – Jan. 2026

🔗🔗 *A tool-augmented agent framework enabling stateful runtime management and code-based function calling.*

- Designed and implemented the persistent Python runtime modules, supporting seamless object injection and retrieval across multi-turn LLM interactions.
- Implemented specific AST-based security rules to validate and sanitize LLM-generated code before execution.
- Developed discovery and registry mechanisms for the Agent Skills open standard, allowing modular injection of functions and types into the runtime.